Report on "Matrix Theory Reloaded: A BPS Road to Holography"

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Summary of the paper

The paper revisits the decoupling limits that define matrix theory. Its starting point is that these are *BPS* limits: they rely on tuning the RR potential so that a Dp-brane's tension is balanced by its charge. Applied to the whole of type II string theory, such a limit gives "matrix p-brane theory" (MpT). Its ten-dimensional target space is non-Lorentzian, with a codimension-$(p+1)$ foliation and a p-brane Galilean boost symmetry. Applied asymptotically to a back-reacted Dp-brane geometry, the same limit induces the near-horizon limit in the bulk. This reproduces $\mathrm{AdS}_5 \times S^5$ and the Itzhaki–Maldacena–Sonnenschein–Yankielowicz (IMSY) geometries. Applying two limits in succession gives new, non-Lorentzian holographic pairs. Uplifting to M-theory, each BPS limit becomes a discrete light-cone quantisation (DLCQ). The authors conjecture that holography is a DLCQ^n/DLCQ^m correspondence with $m > n$. Finally, they read the limit in reverse as a p-brane generalisation of the $T\bar{T}$ deformation. This yields explicit flow equations from free scalars to the Dirac–Nambu–Goto action for all p. It also yields flow equations from scalar plus gauge-field theories to the Dp-brane DBI action, in full for $p = 1, 2$ and to linear order in the deformation parameter for $p = 3$.

1 Principal achievement

The main achievement is a single geometric mechanism that links three areas that are usually treated separately: matrix theory, AdS/CFT and $T\bar{T}$-type deformations. The paper shows that the near-horizon limit is not an extra input. It is the image of the matrix-theory BPS limit applied at infinity. This explains how the non-Lorentzian geometry seen by the worldvolume SYM theory sits inside a Lorentzian AdS bulk, since the AdS metric approaches an MpT-type geometry at large r. The same mechanism is systematic enough to generate new examples, namely the double-limit, non-Lorentzian near-horizon geometries. It also organises both old and new dualities under the DLCQ^n/DLCQ^m classification. The by-products are concrete and checkable. They are the p-brane $T\bar{T}$ flow equations, including a neat three-dimensional equation that applies to both the D2-brane and M2-brane, and an explanation of why the ordinary $T\bar{T}$ flow does not take two-dimensional SYM to the D1 DBI action.

2 Principal drawback

The central claims are structural and conjectural rather than established. The DLCQ^n/DLCQ^m proposal is motivated by a pattern in worked examples (single-brane stacks and the D1–D5 system) and is not derived or independently tested. The new non-Lorentzian holographic backgrounds are constructed by hand, by rescaling the harmonic-function length scale ℓ with the limit parameter. The authors state that they did not check that these backgrounds solve the equations of motion and torsional constraints of the MpT limit of supergravity. They also say that their supersymmetry, charges and physical relevance are not understood. In addition, the non-Lorentzian description at asymptotic infinity may be coordinate dependent. Quantum and

strong-coupling effects become important there, and the $p > 3$ cases need dual descriptions. The $T\bar{T}$ part is likewise incomplete. The flow equations are classical, bosonic and abelian. The $p = 3$ equation is only given to linear order, and the case $p > 3$ is only partially treated. The paper does not show that the deformation makes sense for the non-abelian theories of interest, such as SYM. No holographic observable (correlators, thermodynamics, spectra) is computed in the new examples, so the dictionary remains untested.

3 Most promising follow-up idea

Weighing feasibility against impact, I would pursue **putting the non-Lorentzian holographic backgrounds on a firm footing**. This means formulating the MpT limit of supergravity, which is the equations of motion plus torsional constraints, and verifying that the double-limit near-horizon geometries solve them. The next step is to compute their isometry algebras, supersymmetry and charges, and then one simple holographic observable.

- *Feasibility.* The inputs already exist. The non-relativistic string and M2-brane supergravity equations are available from beta-function and non-Lorentzian bootstrap work, and the paper itself points to the M2 case as a template. The first step is a finite, well-defined computation. It does not require new non-perturbative control: one expands the type II equations in the limit parameter and checks the candidate backgrounds, which are explicit harmonic-function solutions. Isometries and Killing spinors can then be computed by standard methods.
- *Impact.* Without this check the headline "non-Lorentzian holography" examples remain unverified, and so does the DLCQ^n/DLCQ^m conjecture they support. A positive result would make them a controlled top-down laboratory for non-relativistic and Galilean holography. Simple observables could then be computed on both sides, such as thermodynamics via finite-temperature solutions and Wilson-loop or probe-brane spectra. A negative result would be equally useful, because it would show which smearing or localisation choices are physical.

A natural second project, with higher risk and potentially higher reward, is the non-abelian extension of the p-brane $T\bar{T}$ flows, for example M2-brane flows applied to ABJM/BLG theory. The full $p = 3$ flow equation to all orders is a more modest and more feasible task. Both depend less on the holographic checks above and are better treated afterwards.